American Computer Science League

2024-2025 ● Contest 3: Manhattan Distance ● Junior Division

PROBLEM: In a *Right Triangle* pattern, numbers are written in rows that increase in length. The amount by which the rows increase in length is called the pattern's *delta*. In the following example, the delta is 2:

```
 1
 2  3  4
 5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
26 27 28 29 30 31 32 33 34 35 36
....
```

The *Manhattan Distance* between two numbers, *M* and *N*, in a Right Triangle pattern is how far the numbers are from each other in both the horizontal and vertical directions. You'll be given a delta value of a Right Triangle pattern, and two numbers, *M* and *N,* that are in the triangle. Your program must find the Manhattan Distance between the numbers *M* and *N*.

EXAMPLE:

Input	`2 17 3`
Output	`4`
Explanation	The delta value for the Right Triangle is 2; it is shown above. The Manhattan Distance between the numbers 17 and 3 is 4: go horizontally 1 column and vertically 3 rows to go from one number to the other number.

INPUT: Three positive integers, *delta*, *M*, and *N*, each less than or equal to 1,000.

OUTPUT: Find the Manhattan Distance between *M* and *N* in the Right Triangle with the specified delta value.

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SAMPLE INPUT ***(****for readability, all inputs are shown on a single line****)*****:**

```
1. 2 17 3
2. 4 24 25
3. 4 23 7
4. 5 10 50
5. 7 100 5
6. 3 200 150
```

SAMPLE OUTPUT:

```
1. 4
2. 1
3. 8
4. 15
5. 25
6. 20
```

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TEST DATA

TEST INPUT *(for readability, all inputs are shown on a single line):*

```
1. 4 3 45
2. 3 65 17
3. 5 175 200
4. 6 100 185
5. 1 990 1000
6. 2 1000 500
```

TEST OUTPUT:

```
1. 18
2. 12
3. 17
4. 7
5. 35
6. 32
```

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PROBLEM STATEMENT:
In a *Right Triangle* pattern, numbers are written in rows that increase in length. The amount by which the rows increase in length is called the pattern's *delta*. In the following example, the delta value is 2:

```
 1
 2  3  4
 5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
25 27 28 29 30 31 32 33 34 35 36
....
```

The *Manhattan Distance* between two numbers, *M* and *N*, in a Right Triangle pattern is how far the numbers are from each other in both the horizontal and vertical directions. You'll be given a delta value of a Right Triangle pattern, and two numbers, *M* and *N,* that are in the triangle. Your program must find the Manhattan Distance between the numbers *M* and *N*.

EXAMPLE:
Input:
2 17 3

Output:
4

Explanation:
The delta value for the Right Triangle is 2; it is shown above. The Manhattan Distance between the numbers 17 and 3 is 4: go horizontally 1 column and vertically 3 rows to go from one number to the other number.

TASK:
Complete the function **findManDist**

- The function has 3 parameters, all positive integers: *delta*, the increase in the width of each row in the triangle and the two numbers, *M* and *N*, representing the 2 numbers to use in finding the Manhattan Distance.
- The function returns an integer for the Manhattan Distance between where *M* and *N* are in the resulting triangle.

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You may create additional functions that are called from **findManDist** if needed in solving the problem.

CONSTRAINTS:
Each input value is no more than 1,000.

DATA PROVIDED:
There are 6 sets of Sample Data for debugging and 6 sets of Test Data for scoring. The test cases will vary in difficulty. You should create sample data of your own to fully test your program.

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问题：在*直角三角形*模式中，数字按行书写，且每行的长度逐渐增加。每行长度增加的幅度称为该模式的*增量*。在以下示例中，增量为 2：

```
 1
 2  3  4
 5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
25 27 28 29 30 31 32 33 34 35 36
....
```

在直角三角形模式中，两个数字 M 和 N 之间的*曼哈顿距离*是指这两个数字在水平和垂直方向上的距离之和。题目会给出一个直角三角形模式的增量值以及三角形中的两个数字 M 和 N。你的程序需要计算数字 M 和 N 之间的曼哈顿距离。

样例：

输入	2 17 3
输出	4
解释说明	该直角三角形的增量值为 2，如上述示例所示。数字 17 和 3 之间的曼哈顿距离为 4：从 17 到 3 需要水平移动 1 列，垂直移动 3 行。

输入：三个正整数：*增量*、M 和 N，每个数均小于或等于 1000。

输出：在指定增量值的直角三角形中，计算 M 和 N 之间的曼哈顿距离。

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样例输入（*为便于阅读，所有输入均显示在同一行*）：

```
1. 2 17 3
2. 4 24 25
3. 4 23 7
4. 5 10 50
5. 7 100 5
6. 3 200 150
```

样例输出：

```
1. 4
2. 1
3. 8
4. 15
5. 25
6. 20
```

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测试数据

测试输入（*为便于阅读，所有输入均显示在同一行*）：

```
1. 4 3 45
2. 3 65 17
3. 5 175 200
4. 6 100 185
5. 1 990 1000
6. 2 1000 500
```

测试输出：

```
1. 18
2. 12
3. 17
4. 7
5. 35
6. 32
```

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问题陈述：

在*直角三角形*模式中，数字按行书写，且每行的长度逐渐增加。每行长度增加的幅度称为该模式的*增量*。在以下示例中，增量为 2：

```
 1
 2  3  4
 5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
25 27 28 29 30 31 32 33 34 35 36
....
```

在直角三角形模式中，两个数字 M 和 N 之间的*曼哈顿距离*是指这两个数字在水平和垂直方向上的距离之和。题目会给出一个直角三角形模式的增量值以及三角形中的两个数字 M 和 N。你的程序需要计算数字 M 和 N 之间的曼哈顿距离。

***样例*：**

输入：

2 17 3

输出：

4

解释说明：

该直角三角形的增量值为 2，如上述示例所示。数字 17 和 3 之间的曼哈顿距离为 4：从 17 到 3 需要水平移动 1 列，垂直移动 3 行。

任务：

完成函数 findManDist

- 该函数有 3 个参数，均为正整数：*增量*表示三角形中每行宽度的增加值，M 和 N 表示用于计算曼哈顿距离的两个数字。
- 函数返回一个整数，表示在生成的三角形中， M 和 N 之间的曼哈顿距离。

在解决问题的过程中，如有需要，你可以构建额外的函数供 findManDist 调用来解答此问题。

约束条件：
每个输入值不超过 1000。

提供的数据：
有 6 组用于调试的样例数据和 6 组用于评分的测试数据。测试用例的难度各不相同。你应该创建自己的样例数据来充分测试你的程序。